

Lab 12 Template

Some questions require multiple parts to be answered, be sure to discuss them in full.

Part 01:

1. **On which days** were you in contact with the infected individual? **Take a screenshot** of the location and time where you were in contact with the COVID-19 patient to include in your lab report.

(10 points)

```
cpre3310@cpre3310:~/lab12/part01$ python3 part01_patient_tracing_skel.py
Common hashes found: ['16af42a4fa13e6706145d996076642441c7fff5e729ebed2be5b33602b05b1b8', '75427abfb6bc8293015ba0a23dbc71b054cc06cfd77d9ce6447914b54aa686e']
Common locations corresponding to those hashes: [{'location': 'MUPandaExpress', 'time': '2025-11-11T12:30:00'}, {'location': 'Coover1041LabA', 'time': '2025-11-12T16:30:00'}]
cpre3310@cpre3310:~/lab12/part01$
```

By looking at the screenshot, we can say that we met the infected person on 2025-11-11 12:30:00 at MUPandaExpress and 2025-11-12 16:30:00 at Coover1041LabA.

2. **What do you know about the infected person?**

(5 points)

I basically know that the infected person's hashed movement was compared against mine and basically from the PSI process, I can only learn about matching the contact events, but I do not learn about the patient's identity, name or about their full movement history.

3. **Identify a flaw with the use of hashing in this scenario.** Consider how errors and inaccuracies in time and location might impact the hash.

(10 points)

In this case I feel that even if the recorded time or the location is slightly different, the hash will be very different and no match will be found, which means that even small errors or inaccuracies in time and location data can cause the real contacts to be missed.

4. Please provide **another real-world example where PSI could be beneficial**. Justify your reasons why it maintains **confidentiality** and **privacy** in your proposed use case. Explain how it benefits **public good** and protects **individual rights**.

(25 total points, 5 each set of bold words)

Another real world, example that we can use over here is of the fraud detection between banks where different banks may want to identify customers or accounts that are connected to the stolen identities, money laundering, or repeated

fraudulent transactions; however, they cannot simply share their entire customer databases since that could expose some private and sensitive financial information. Now, with PSI, each bank can compare its list of suspicious accounts with another bank's list and learn only about the accounts that appear in both datasets, which helps in maintaining confidentiality since both banks may keep rest of their records private and also protects privacy since only the overlapping suspicious entries are revealed instead of everyone's personal financial information, which benefits the public good by helping financial institutions that detect fraud more effectively, reduce the financial crime and help in improving the trust in banking systems and it also protects individuals rights since innocent customers do not have their personal information shared without a valid reason.

Part 02:

5. **Upload** your code to Canvas.
(10 points)

Part 03:

6. **Take a screenshot of the 4 plaintext averages.**
(10 points)

```
(phe_env) cpre3310@cpre3310:~/lab12/part03$ python3 part03_drug_trial_researcher_skel.py  
Decrypted averages: [126.25, 82.5, 61.25, 73.25]  
(phe_env) cpre3310@cpre3310:~/lab12/part03$
```

7. **Upload** your code to Canvas.
(5 points)
8. **How could partially homomorphic encryption be effectively used in an election?** 3 viewpoints
(25 total points, 8.33 points each perspective)

A partially homomorphic encryption could be used be effectively in an election since it allows the votes to be remained encrypted while still being added together to produce a final tally, like for example, each voter's ballot could be encrypted as soon as it is cast and instead of storing or counting the plaintext directly, the system would basically store encrypted votes and use homomorphic addition to combine them and at the end of the elections, the final totals would be decrypted by authorized officials, which helps us in protecting ballot secrecy while still allowing accurate vote counting.

From the poll workers' viewpoint: The poll workers should not be able to see the plaintext vote for a candidate since their job is to help manage the polling

place, check in voters, and make sure the election procedures are followed, so if partially homomorphic encryption is used, the poll worker would only see that a ballot was successfully submitted and encrypted, not to the selected voter, which protects the voter privacy and basically reduces the risk of coercion, intimidation, or bias at the polling place. The poll workers can still confirm that the machine is working and each voter casts only 1 ballot, but they should not have access to the actual vote contents.

From the vote counters' viewpoint: The vote counters should not be able to see individual plaintext ballots since their role is to total the votes and verify accuracy and with partially homomorphic encryption, they can basically add encrypted votes together without actually decrypting each one separately, which means that they can produce an encrypted tally for each candidate while preserving the secrecy of individual ballots, so if decryption is needed, it should only be used on the final totals instead of using it on each person's vote, which makes recounts and audits safer because the system can recheck totals without exposing how any one person voted.

From the election officials' perspective: The election officials should not be able to see the individual plaintext voters as well, since they only have the access to the final decrypted results, such as the total number of votes each candidate received, and possibly summaries of some group totals from some polling districts if its required for reporting. Also, their responsibility is to collect results from polling places, verify integrity, and basically certify the election and using partially homomorphic encryption basically helps them report accurate outcomes while protecting the ballot confidentiality and it also strengthens the public trust because the election can be audited and certified without revealing individual voter choices.

Overall, we can say that homomorphic encryption is effective in elections because it balances accuracy, privacy and trust. The plaintext vote must remain hidden from poll workers, vote counters and the election officials at the individual level and only the authorized decryption if final totals should occur, since it protects voters while still showing the election to be completed and verified correctly.